Assignment 2: Registration System Test Report

Requirements-based black-box and white-box testing, defect evidence, and regression results

1. Scope and execution

The supplied Student Course Registration System was treated as the system under test. Expected results derive from REQUIREMENTS.md (FR-01 through FR-15) and were not adjusted to match implementation behavior. The source was inspected during the initial repository inventory before test code was written, so the recommended ordering was not followed exactly; baseline execution still used the unmodified starter. Tests run with Python 3.12, pytest 9.2.0, and pytest-cov 7.1.0.

2. Black-box design

Equivalence class partitioning covers valid/invalid current credits and course credits; zero, positive, and negative dues; empty/all/some/none prerequisite completion; valid seat states and invalid capacity/enrollment. The workbook contains 23 traceable input and decision cases, exceeding the required eight ECP cases.

Boundary analysis includes current credits 0 and 18; new course credits 1 and 4 plus invalid 0 and 5; combined total 18 versus 19; seats at capacity-1 and capacity; and dues Rs. 0 versus Rs. 1. The executable black-box suite has 35 collected cases across ECP, BVA, decision outcomes, exceptions, and state changes.

Decision table and priority

Rule	Conditions in priority order	Expected status	Executable trace
R1	dues > 0	DUES_BLOCK	DT-PRIORITY-DUES; BVA-DUES-1
R2	dues=0; duplicate	ALREADY_REGISTERED	DT-PRIORITY-DUP
R3	dues=0; no duplicate; prerequisite missing	PREREQUISITES_MISSING	DT-PRIORITY-PREREQ
R4	prior clear; credit total > 18	CREDIT_LIMIT	DT-PRIORITY-CREDIT
R5	prior clear; enrolled=capacity	WAITLISTED	BVA-CAPACITY-FULL
R6	prior clear; enrolled<capacity	REGISTERED	DT-REGISTER
Invalid	negative dues or invalid seats; validation precedes statuses	ValueError (FR-15)	ECP-DUES-NEGATIVE; ECP-CAPACITY-ZERO; ECP-ENROLLMENT-*

Optimization: no exact duplicates were found across ECP, BVA, and decision-table cases. Similar outputs were retained where inputs, requirement intent, state effects, or priority combinations differ. Technique-to-test traceability is maintained in the workbook and test IDs.

3. Baseline results and defect evidence

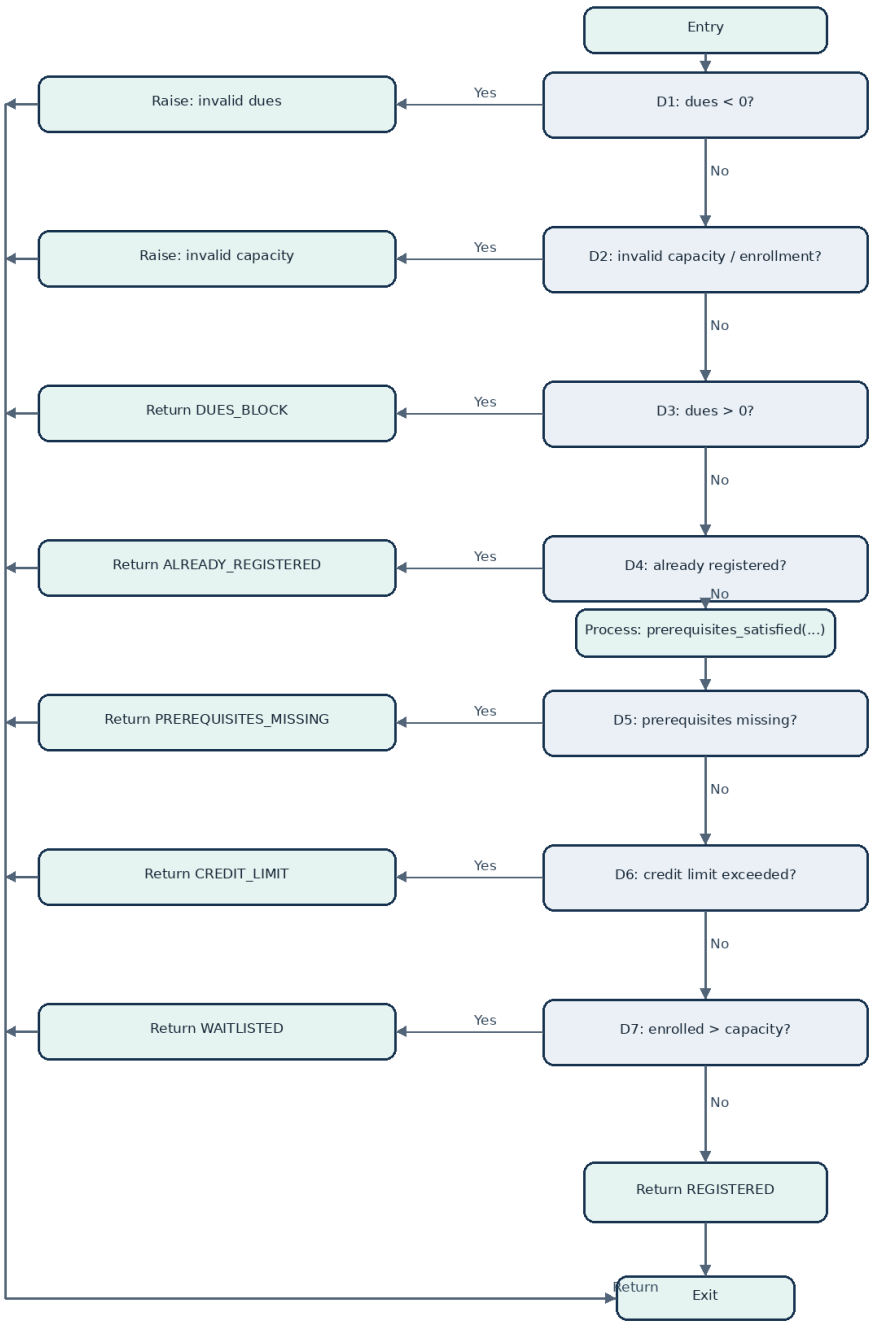
Before source fixes, the black-box suite produced 11 failures and 24 passes (35 collected cases). Failures were retained as legitimate evidence; expected values were not changed. The 11 failures expose five distinct defects, with multiple cases confirming some defects.

Defect	Requirement	Reproduction / expected vs starter actual	Failed test ID
D-01	FR-01	15 + 3 credits should be True at total 18; starter returned False.	BVA-TOTAL-18
D-02	FR-03 / FR-06	Only CS101 completed of CS101 and MATH101 should fail prerequisites; starter returned True.	ECP-PR-SOME
D-03	FR-04 / FR-09	Rs. 1 dues should return DUES_BLOCK; starter returned REGISTERED.	BVA-DUES-1
D-04	FR-08	At capacity should return WAITLISTED; starter returned REGISTERED. Over-capacity is rejected earlier.	BVA-CAPACITY-FULL
D-05	FR-13	Withdrawal should decrement course enrollment; starter left enrollment unchanged.	STATE-WITHDRAW

The 1001-rupee positive-dues ECP case alone does not expose D-03 because the starter blocks only balances above 1000. The Rs. 1 BVA case isolates the requirement boundary. The separate defects log includes reproduction steps, expected/actual results, failed test IDs, and fix status.

4. Control-flow graph and cyclomatic complexity

CFG convention: each if statement is one decision node. The compound capacity validation on line 48 and chained comparison in check_credit_limit are each treated as one decision for the registration_decision CFG; their internal Python short-circuit conditions are expanded separately in the MC/DC discussion. The prerequisites_satisfied call is shown as a processing node and is not expanded into this function graph.



Source: registration_decision(); short-circuit subexpressions counted as one decision per if statement.

Nodes N=18; edges E=24; V(G)=E-N+2=24-18+2=8. There are seven if decision nodes, so decision nodes + 1 = 7+1=8. The calculations agree.

Independent basis paths

Path	Representative test	Decision outcome
P1	WB-EXC-ORDER	Invalid dues exception
P2	WB-CAP-A	Invalid capacity exception
P3	DT-PRIORITY-DUES	DUES_BLOCK
P4	DT-PRIORITY-DUP	ALREADY_REGISTERED
P5	DT-PRIORITY-PREREQ	PREREQUISITES_MISSING
P6	DT-PRIORITY-CREDIT	CREDIT_LIMIT
P7	BVA-CAPACITY-FULL	WAITLISTED (reaches this path after D-04 fix)
P8	DT-REGISTER	REGISTERED

Each path takes a distinct first decision outcome edge, with P8 taking every status decision false. WB-CAP-A is the invalid-enrollment path; the second invalid-seat tests exercise the other short-circuit outcomes.

5. Condition coverage and MC/DC

Selected compound decision: check_credit_limit current-credit validation, logically A AND B where A is current_credits >= 0 and B is current_credits <= 18. Python evaluates the chained comparison left-to-right; if A is false, B is not evaluated.

Test / input	A	B evaluated? / value	Decision
ECP-CR-CURRENT-LOW: current=-1	False	Not evaluated	Invalid current credits
ECP-CR-VALID-MIN: current=0	True	True	Continue validation
ECP-CR-CURRENT-HIGH: current=19	True	False	Invalid current credits
BVA-TOTAL-19-AT-MAX: current=18	True	True	Course addition exceeds maximum

MC/DC pairs: A is independently shown by current=-1 versus current=0 (the logical decision changes; B is short-circuited for -1 and would be True if evaluated). B is independently shown by current=18 versus current=19 while A remains True. The two pairs account for Python short-circuiting; pytest-cov does not calculate condition coverage or MC/DC.

6. Coverage analysis

Run	Statement coverage	Branch coverage	Uncovered / action
Initial black-box suite on starter	55/57 = 96.5%	24/26 = 92.3%	Lines 59 (WAITLISTED return) and 71 (waitlist append); 2 missing branches.
After white-box additions on starter	56/57 = 98.2%	25/26 = 96.2%	Line 59 and its incoming decision edge remain unreachable under FR-15 seat validation. WB-WAITLIST-ADD and WB-WAITLIST-DUP cover line 71 and both waitlist membership edges via isolated decision stubs.
After fixes, all tests	58/58 = 100%	26/26 = 100%	No uncovered statements or branches.

Starter coverage JSON is preserved at docs/coverage_initial.json and docs/coverage_whitebox.json; final coverage is in docs/coverage_final.json. Coverage includes statement and branch metrics only. The full post-fix command completed with 42 passed.

7. White-box test targets

Test ID	Target
WB-CAP-A	Capacity validation: negative enrollment; first chained comparison fails and short-circuits.
WB-CAP-B	Capacity validation: enrollment above capacity; second chained comparison fails; OR right side skipped.
WB-CAP-C	Capacity validation: (capacity,enrolled)=(0,0); chained comparisons pass and capacity<1 is evaluated.
WB-EXC-ORDER	FR-15: negative dues exception precedes invalid capacity exception.
WB-CAP-VALID	False edge from seat-validation decision for valid enrollment.
WB-WAITLIST-ADD	register_course WAITLISTED branch; stub decision to reach waitlist append integration branch.
WB-WAITLIST-DUP	register_course WAITLISTED branch; existing student avoids duplicate append.

8. GitHub collaboration and submission links

Repository: https://github.com/17sync/Application-Defect-Management

Branches: black-box (requirements-based tests and preserved failures), white-box (CFG, coverage and structural tests), fixes (implementation corrections and passing regressions). Main remains unmodified.

Issues: #1 capacity/waitlist https://github.com/17sync/Application-Defect-Management/issues/1; #2 prerequisites https://github.com/17sync/Application-Defect-Management/issues/2; #3 dues https://github.com/17sync/Application-Defect-Management/issues/3; #4 withdrawal https://github.com/17sync/Application-Defect-Management/issues/4; #5 exact credit maximum https://github.com/17sync/Application-Defect-Management/issues/5.

Pull request references: PR URLs will be listed after the remote branches are published. Peer review attribution requires an actual second team member; no member names were supplied, so no reviewer activity is invented.

Contributions: repository author identity is the configured Git identity; additional team member contributions must be recorded by the team.

The repository includes the black-box case workbook, decision table workbook, CFG image, test suites, defect log, and this report. GitHub Actions workflow runs pytest with branch coverage on pushes and pull requests and uploads coverage.xml even after a failed run.